ggplot2 internals · aesthetics · geoms

Day 7 - Wednesday, May 27, 2026

> ℹ Today: Wednesday May 27
>
> **Before class**
>
> - **Perusall Assignment**: DC §6.1–6.5
>
> **Plan for today**
>
> - The grammar of graphics: a vocabulary for *any* plot
> - Aesthetic mappings vs. fixed properties (and a common bug)
> - Geoms in depth: layering, local vs. global, multiple data
> - Statistical transformations: where do bar heights come from?
> - Position adjustments: stacking, dodging, jittering
> - In-class activity
>
> **Helpful links**
>
> - Syllabus · AI policy · Schedule · *R4DS* Ch. 9 · DC §6 · ggplot2 reference

Quick recap

We've been making ggplots for a while now. You know how to:

- Pass `data` and `aes()` to `ggplot()`
- Add a geom like `geom_point()`, `geom_bar()`, `geom_histogram()`
- Map variables to `x`, `y`, `color`, `fill`

Today we're going to **slow down** and look at what's actually happening under the hood.

Tip

The goal of today is to give you the tools to be able to build any plot you can describe.

The grammar of graphics

A vocabulary for plots

The DC chapter introduces five words. These will come up a lot.

Table 1: Five words to describe any data graphic.

Term	What it is
frame	The relationship between *position* and *data*. The coordinate system + the variables on each axis.
glyph	The basic graphical unit. One glyph = one case in your data frame.
aesthetic	Any graphical attribute of a glyph: position, size, color, shape, transparency, …
scale	The rule that translates a *variable value* into an *aesthetic value*.
guide	What helps the viewer translate back: axes, tick marks, legends, color bars.

One sentence, five words

Tip

A graphic is a collection of **glyphs** sitting inside a **frame**, where each glyph's **aesthetics** are controlled by **scales** that map variables to graphical attributes, and **guides** help the viewer read the scales backward.

Today we focus on the first three; tomorrow we'll deal more with scales and guides.

Why this matters

Once you have the vocabulary, debugging gets easier:

- "My color isn't working" → is it a *mapping* (inside `aes()`) or a *fixed property* (outside)?
- "I want different lines per group" → I need a discrete variable mapped to an aesthetic that *splits* the geom (color, linetype, group).

- "My bar chart shows counts but I want proportions" → I need to override the **stat**.

Every confusion you've had with ggplot maps onto one of these concepts.

A dataset for today

The `storms` dataset

Built into the tidyverse (via `dplyr`). Records of every named Atlantic storm since 1975, drawn from NOAA's HURDAT2 reanalysis.

```
storms
```

```
# A tibble: 20,778 x 13
   name   year month   day  hour   lat  long status      category  wind pressure
   <chr> <dbl> <dbl> <int> <dbl> <dbl> <dbl> <fct>          <dbl> <int>    <int>
 1 Amy    1975     6    27     0  27.5 -79   tropical d~       NA    25     1013
 2 Amy    1975     6    27     6  28.5 -79   tropical d~       NA    25     1013
 3 Amy    1975     6    27    12  29.5 -79   tropical d~       NA    25     1013
 4 Amy    1975     6    27    18  30.5 -79   tropical d~       NA    25     1013
 5 Amy    1975     6    28     0  31.5 -78.8 tropical d~       NA    25     1012
 6 Amy    1975     6    28     6  32.4 -78.7 tropical d~       NA    25     1012
 7 Amy    1975     6    28    12  33.3 -78   tropical d~       NA    25     1011
 8 Amy    1975     6    28    18  34   -77   tropical d~       NA    30     1006
 9 Amy    1975     6    29     0  34.4 -75.8 tropical s~       NA    35     1004
10 Amy    1975     6    29     6  34   -74.8 tropical s~       NA    40     1002
# i 20,768 more rows
# i 2 more variables: tropicalstorm_force_diameter <int>,
#   hurricane_force_diameter <int>
```

Variables we'll use

Table 2: Selected variables in `storms`.

Variable	Description
`name`, `year`, `month`, `day`, `hour`	When the observation was taken
`lat`, `long`	Where the storm was
`status`	`tropical depression`, `tropical storm`, `hurricane`, …
`category`	Saffir-Simpson scale (1–5), `NA` for non-hurricanes
`wind`	Maximum sustained wind speed, knots

Variable	Description
`pressure`	Central pressure, millibars (lower = stronger)

Each row is a **6-hour observation** of one storm, so single hurricane shows up many times as it moves and intensifies.

Aesthetic mappings

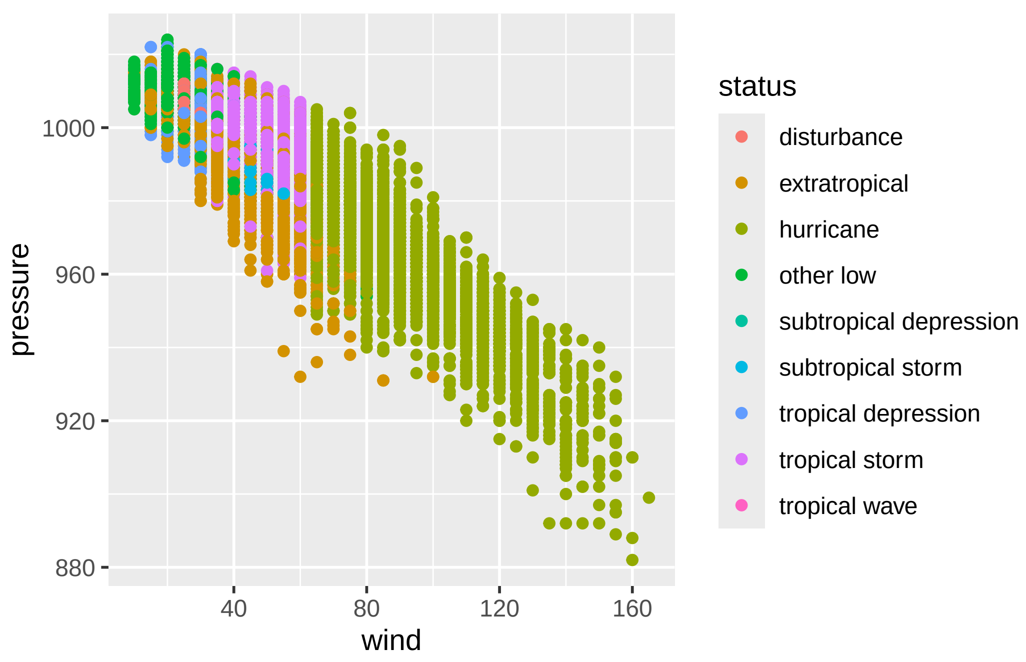
Mapping vs. setting

The single most common ggplot bug is confusing **mapping** with **setting**.

Mapping: a variable controls the aesthetic.

Goes **inside** `aes()`.

```
ggplot(storms,
       aes(x = wind, y = pressure,
           color = status)) +
  geom_point()
```

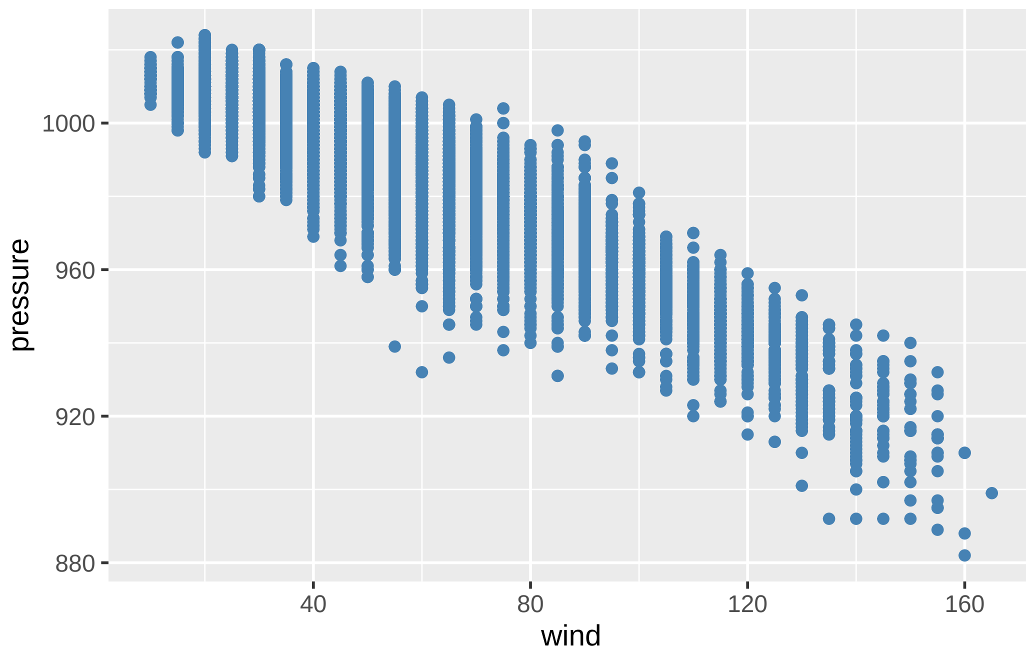


`color` varies by `status` value.

Setting: you fix the aesthetic to a constant.

Goes **outside** `aes()`.

```
ggplot(storms,
       aes(x = wind,
           y = pressure)) +
  geom_point(color = "steelblue")
```

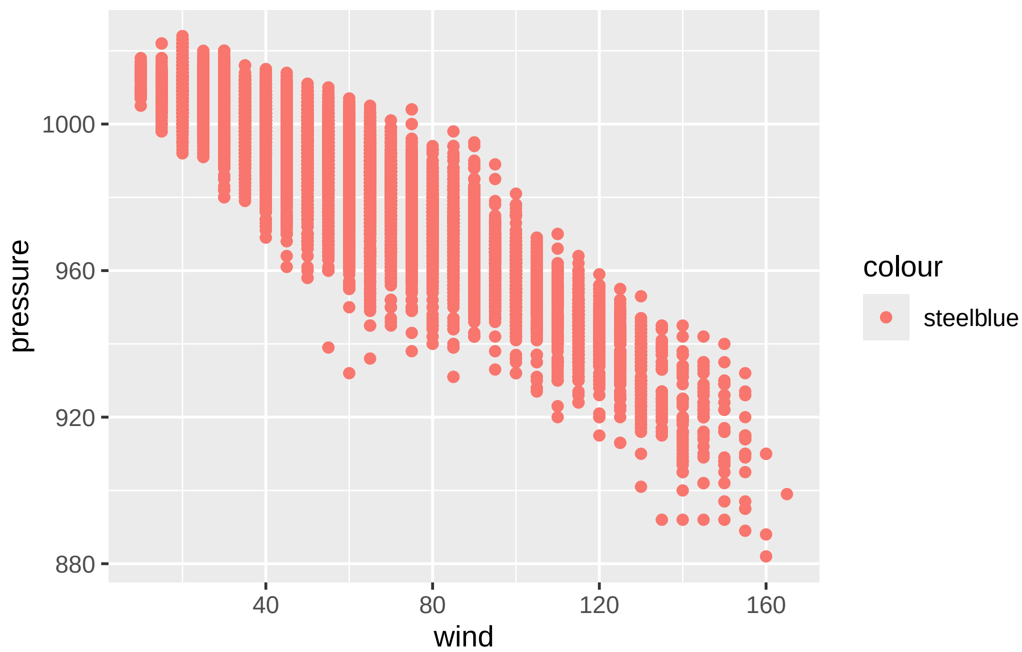


Every point is steelblue.

The classic mistake

What does this do?

```
ggplot(storms, aes(x = wind, y = pressure)) +
  geom_point(aes(color = "steelblue"))
```



It makes every point a **salmon red** and adds a legend that says `"steelblue"`.

Why? `aes()` says "map a variable to color." We gave it the string `"steelblue"`, which ggplot treated as a one-level categorical variable. It then picked a default color (red-pink) for that one level.

Rule of thumb: if you want a constant color, it goes *outside* `aes()`. If you want it to depend on data, it goes *inside* `aes()`.

What can you map?

Every geom accepts a subset of these aesthetics. The most common:

Table 3: Common aesthetics.

Aesthetic	Works on	Notes
`x, y`	almost everything	position in the frame
`color`	points, lines, borders	outline of a glyph
`fill`	bars, areas, filled shapes	interior of a glyph
`size`	points, lines	numerical only, ideally
`alpha`	almost everything	transparency, 0–1
`shape`	points	max 6 discrete values by default
`linetype`	lines	solid, dashed, dotted, …

Aesthetic	Works on	Notes
`group`	grouped geoms	splits w/o adding a legend

Discrete vs. continuous aesthetics

ggplot picks a scale automatically based on the **type** of the variable.

Table 4: How ggplot scales aesthetics by variable type.

Aesthetic	Discrete variable	Continuous variable
`color` / `fill`	qualitative palette (distinct hues)	gradient (e.g., light → dark)
`size`	warns (order is implied but levels aren't ordered)	smooth size range
`shape`	distinct shapes (max 6)	error (shapes don't have an order)
`alpha`	warns, same reason as size	smooth

> ⚠ Warning
>
> Mapping a categorical variable like `status` to `size` or `alpha` tells the reader "these categories are ordered", which is often not actually the case. Stick to color, fill, shape, or facets for categorical splits.

Shape reference

In `ggplot`, you specify the shape of a point as a number.

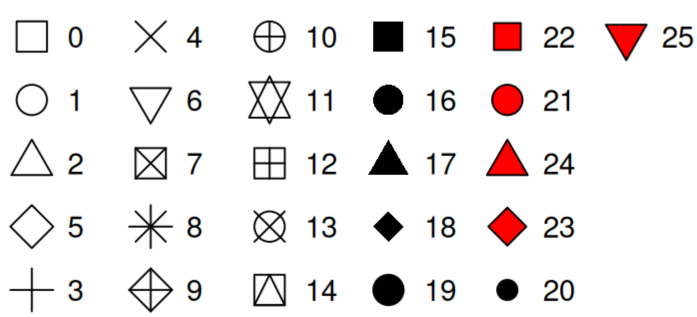
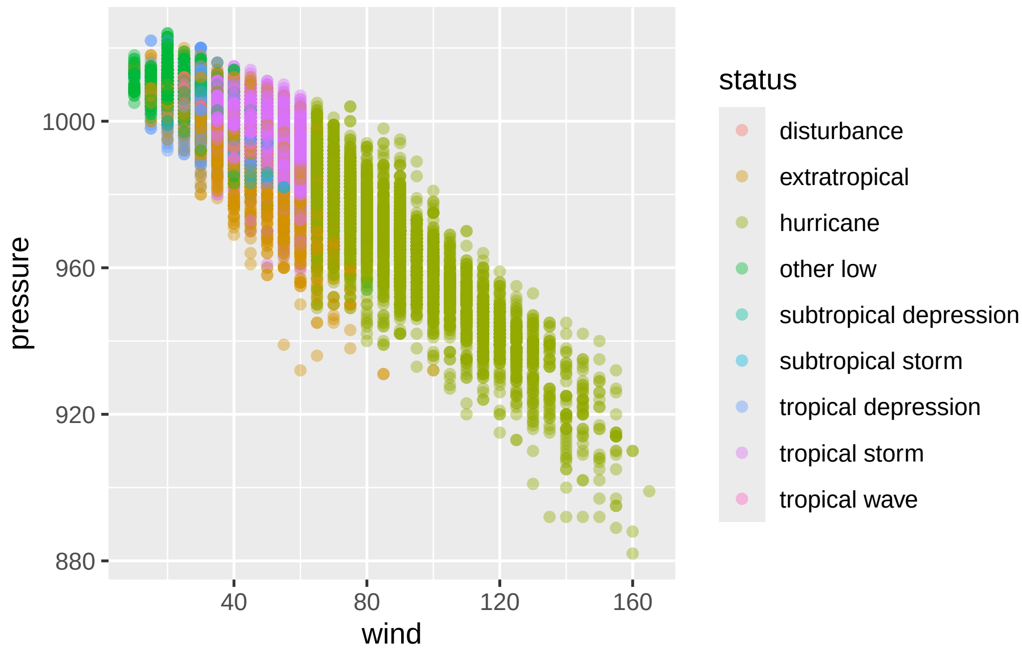


Figure 1: R has 26 built-in shapes that are identified by numbers.

Example: status mapped four ways

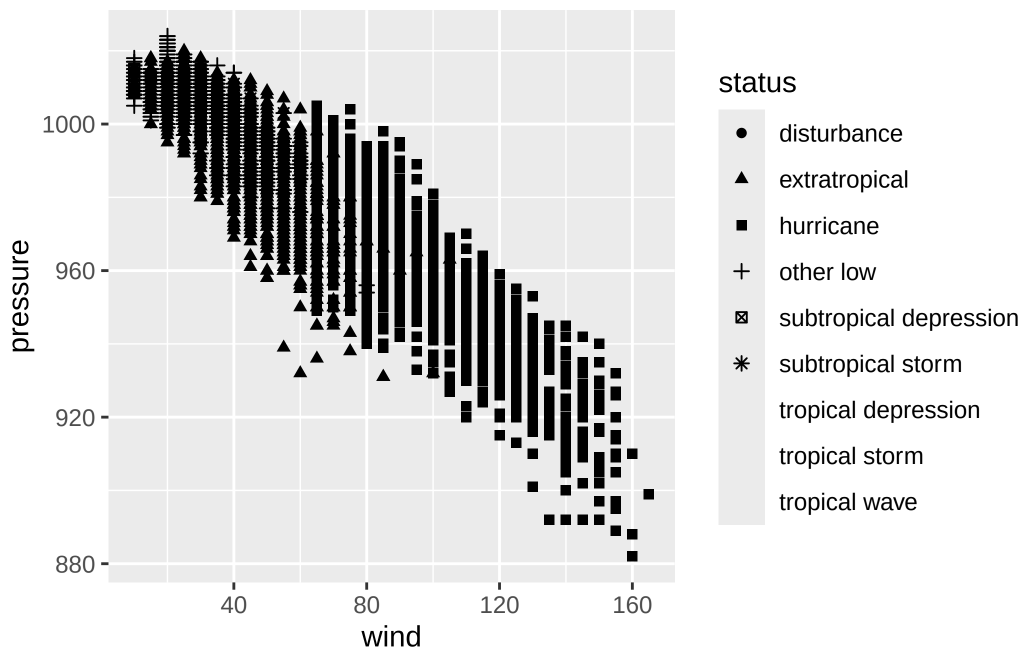
```
# Color: good for categorical
ggplot(storms, aes(x = wind, y = pressure, color = status)) + geom_point(alpha = 0.4)
```




```
# Shape: also good, but watch the 6-value limit
ggplot(storms, aes(x = wind, y = pressure, shape = status)) + geom_point()
```

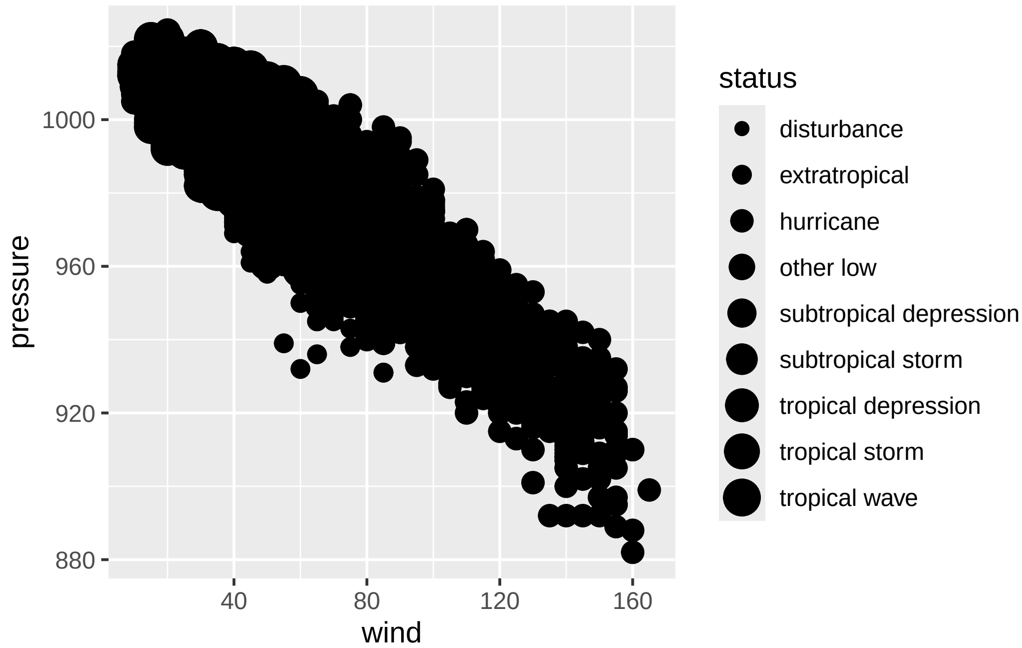
```
Warning: The shape palette can deal with a maximum of 6 discrete values because more
than 6 becomes difficult to discriminate
i you have requested 9 values. Consider specifying shapes manually if you need
  that many of them.
```

```
Warning: Removed 11048 rows containing missing values or values outside the scale range
(`geom_point()`).
```



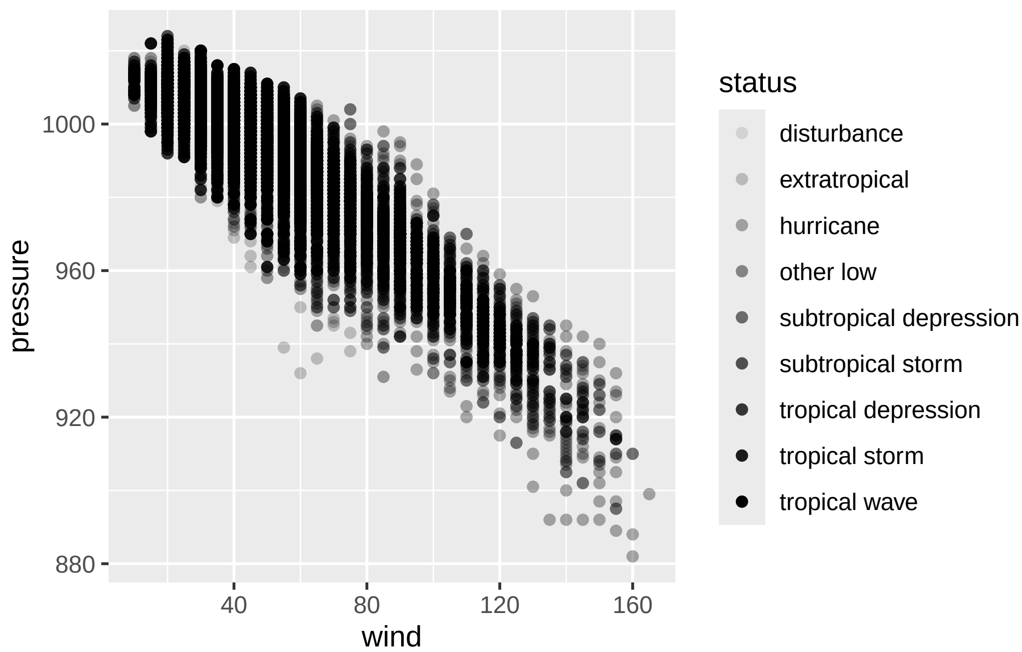

```
# Size: bad: implies ranking
ggplot(storms, aes(x = wind, y = pressure, size = status)) + geom_point()
```

```
Warning: Using size for a discrete variable is not advised.
```




```
# Alpha: same problem as size
ggplot(storms, aes(x = wind, y = pressure, alpha = status)) + geom_point()
```

```
Warning: Using alpha for a discrete variable is not advised.
```

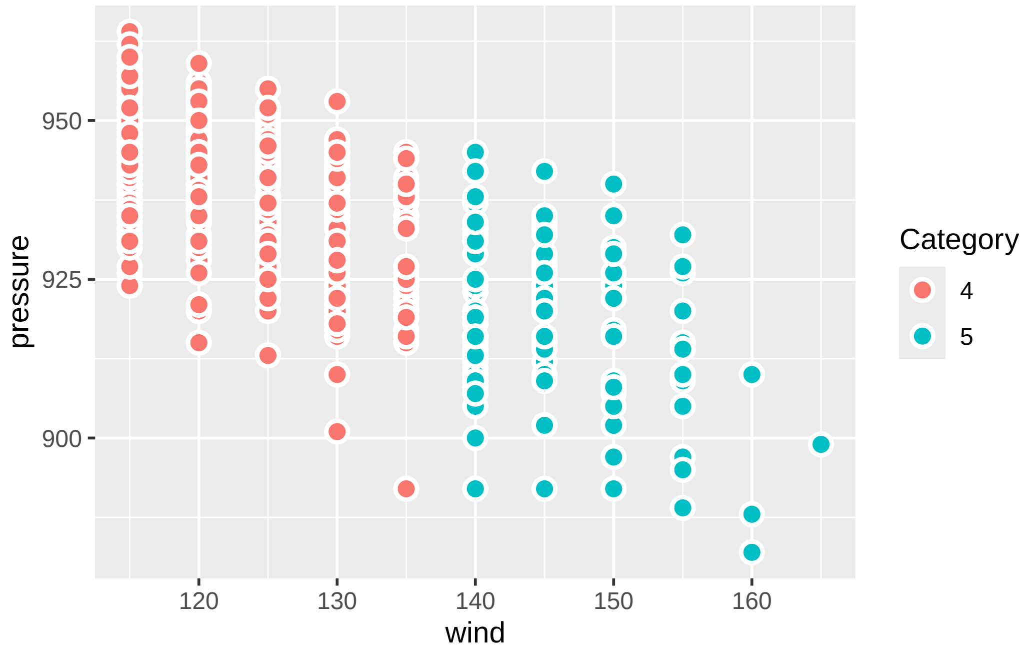


Using size/alpha for a discrete variable is not advised.

The `stroke` aesthetic

For filled point shapes (21–25), `stroke` controls border thickness independently of `size`.

```
ggplot(storms |> filter(category >= 4),
       aes(x = wind, y = pressure, fill = factor(category))) +
  geom_point(shape = 21, size = 3, stroke = 1.2, color = "white") +
  labs(fill = "Category")
```

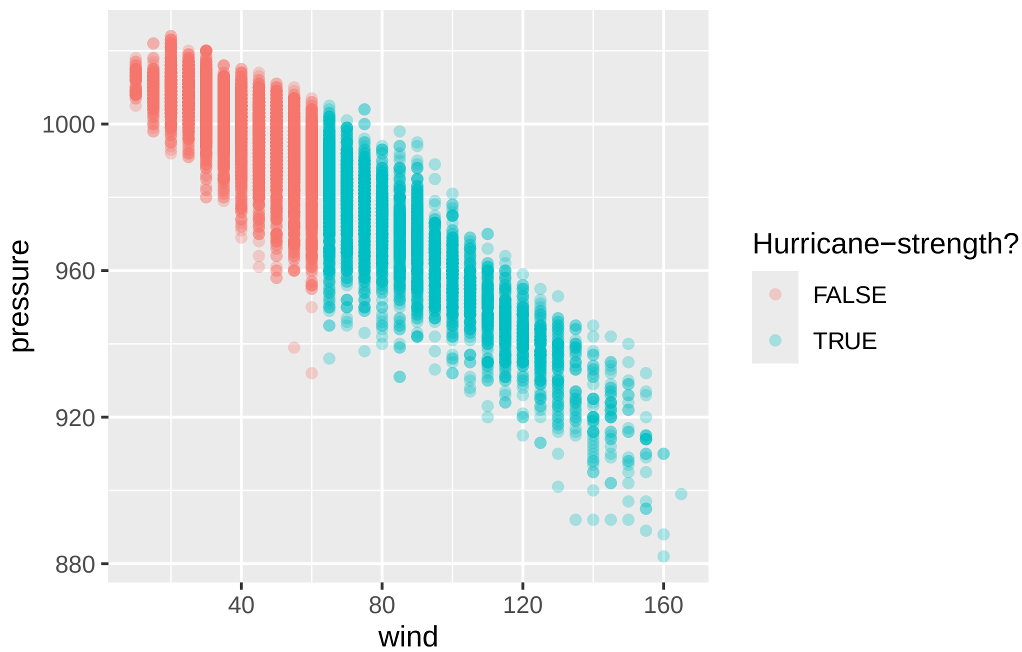


A white stroke around colored points is one of the easiest ways to make a busy plot readable.

On-the-fly transformations

Aesthetics don't have to map to a variable name; they can map to *any expression* involving variables.

```
ggplot(storms,
       aes(x = wind, y = pressure, color = wind >= 64)) +
  geom_point(alpha = 0.3) +
  labs(color = "Hurricane-strength?")
```



The expression `wind >= 64` returns `TRUE`/`FALSE` for each row, and that gets mapped to color. No need to create the column in advance.

Geoms in depth

Geoms = glyph families

Each geom produces one *type* of glyph. The big ones:

Table 5: Common geoms.

Geom	Glyph	Best for
`geom_point()`	dot	scatter
`geom_line()`	connected segments	trends over an ordered x
`geom_smooth()`	one curve (per group)	summarize a trend
`geom_bar()` / `geom_col()`	rectangles	counts / values by category
`geom_histogram()`	rectangles (bins)	distribution of one numeric var
`geom_boxplot()`	boxes + whiskers	distribution by group
`geom_density()`	smooth curve	distribution, smoothed
`geom_polygon()`	filled shapes	maps, custom regions

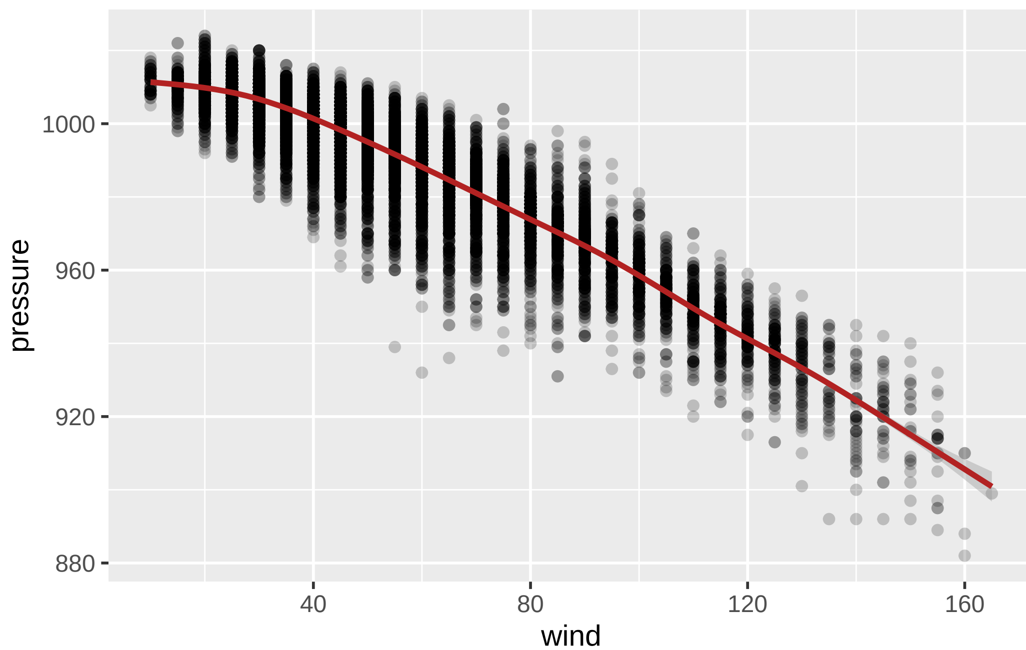
ggplot2 ships with 40+. Extension packages add hundreds more.

Layering: more than one geom

A plot can have as many geoms as you want. They stack in the order you write them.

```
ggplot(storms, aes(x = wind, y = pressure)) +
  geom_point(alpha = 0.2) +
  geom_smooth(color = "firebrick")
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```



Here we have two layers, the same data, and the same mappings. The points show every observation; the smooth shows the trend.

> **Note**
>
> The relationship between wind and pressure in tropical cyclones is *very* strong. The scatter shows what that law looks like in real storm data.

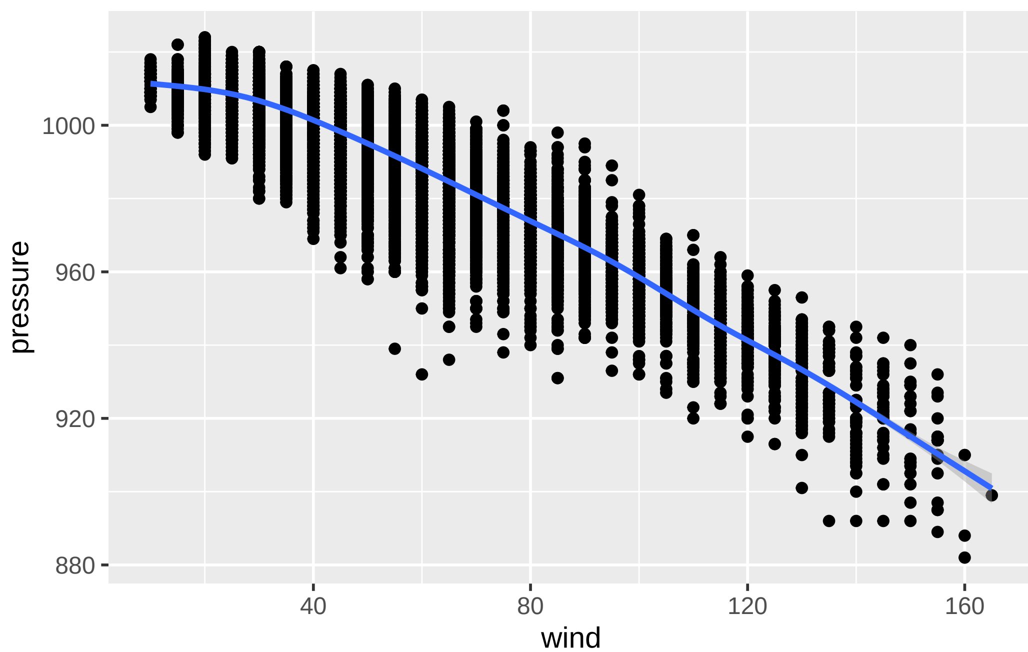
Global vs. local mappings

You can put `aes()` in two places:

In `ggplot()`: global. Every layer inherits these mappings.

```
ggplot(storms,
       aes(x = wind, y = pressure)) +
  geom_point() +
  geom_smooth()
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

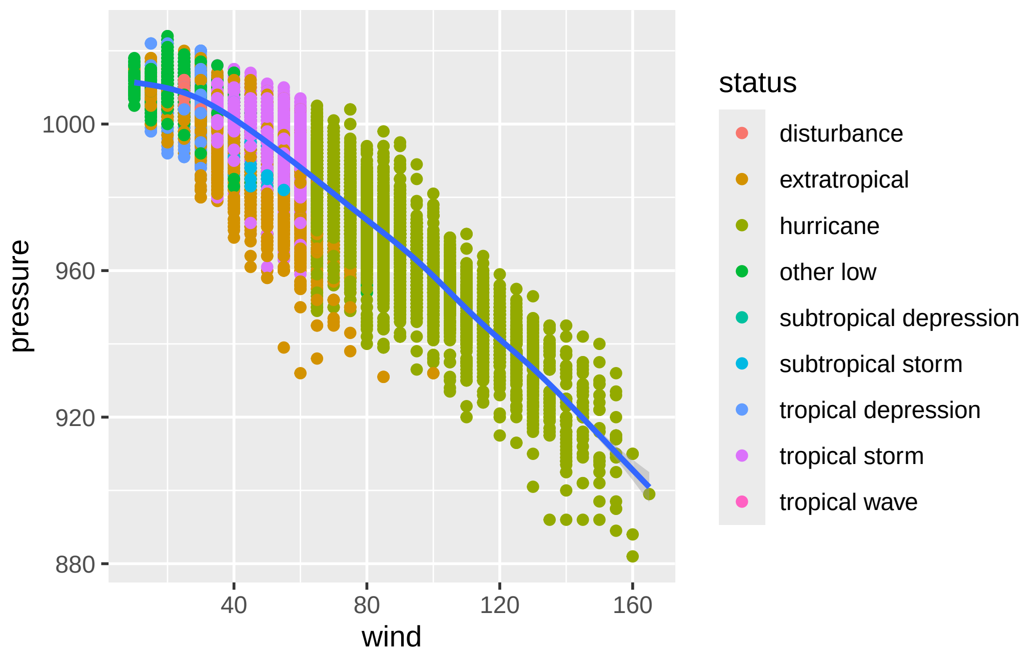


Both geoms see x and y.

In a geom: local. Only that layer sees it.

```
ggplot(storms,
       aes(x = wind, y = pressure)) +
  geom_point(aes(color = status)) +
  geom_smooth()
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```



Only points are colored; the smooth is one curve.

Why local mappings matter

If you put `color = status` in the global `aes()`, **every layer** sees it. `geom_smooth()` will draw one smoother *per status*, not one overall.

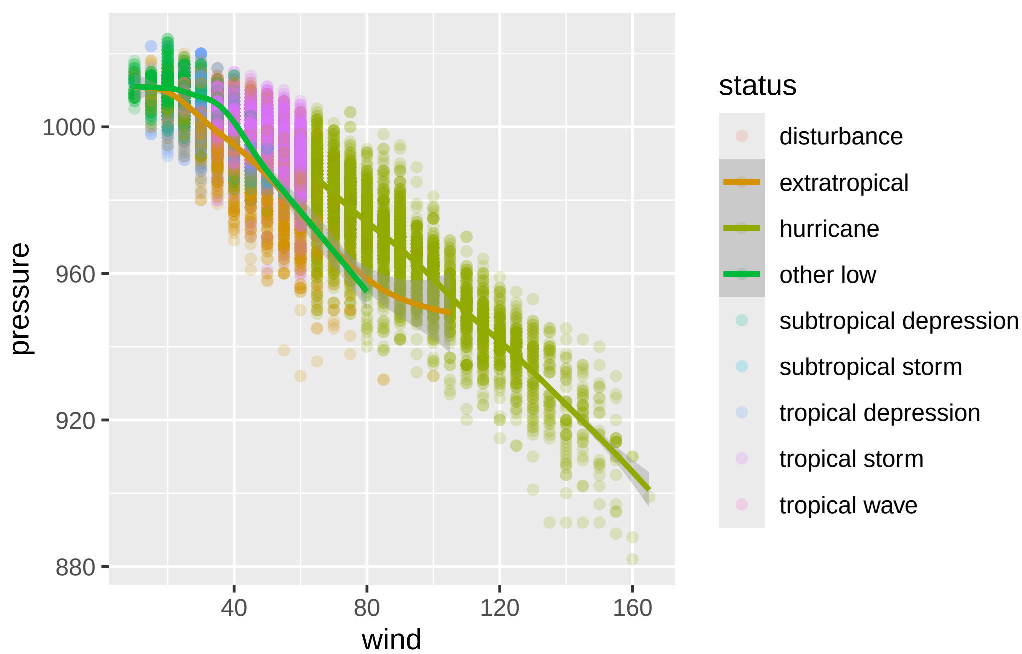
```
# One smoother per status (probably not what you want here):
ggplot(storms, aes(x = wind, y = pressure, color = status)) +
  geom_point(alpha = 0.2) +
  geom_smooth()
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

```
Warning: Failed to fit group 1.
Caused by error in `smooth.construct.cr.smooth.spec()`:
! x has insufficient unique values to support 10 knots: reduce k.
```

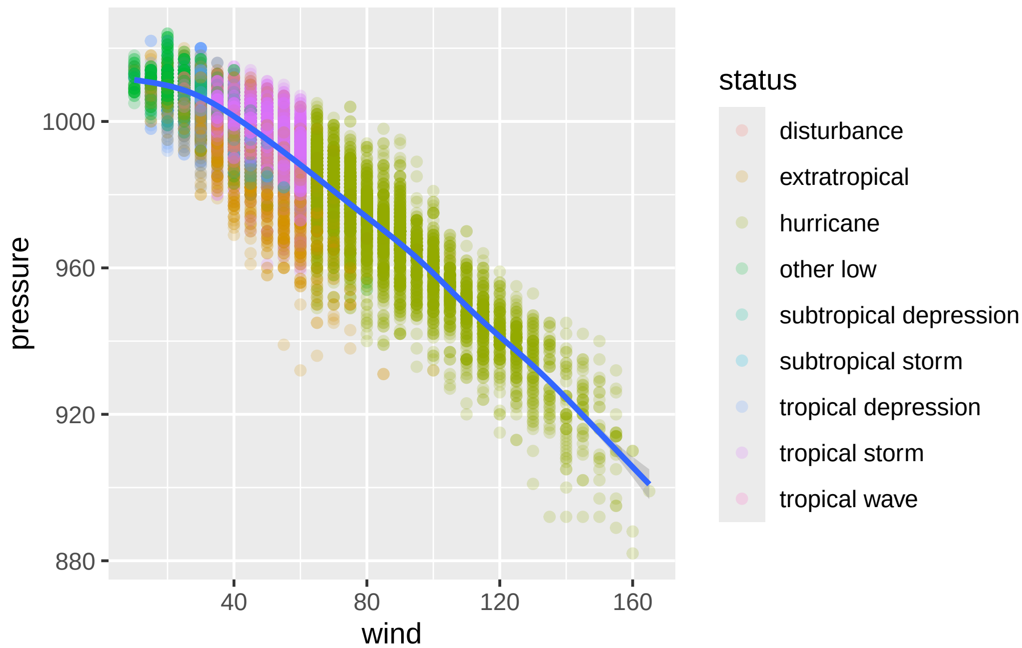
```
Warning: Failed to fit group 5.
Caused by error in `smooth.construct.cr.smooth.spec()`:
! x has insufficient unique values to support 10 knots: reduce k.
```

```
Warning: Failed to fit group 6.
Caused by error in `smooth.construct.cr.smooth.spec()`:
! x has insufficient unique values to support 10 knots: reduce k.

Warning: Failed to fit group 7.
Caused by error in `smooth.construct.cr.smooth.spec()`:
! x has insufficient unique values to support 10 knots: reduce k.

Warning: Failed to fit group 8.
Caused by error in `smooth.construct.cr.smooth.spec()`:
! x has insufficient unique values to support 10 knots: reduce k.

Warning: Failed to fit group 9.
Caused by error in `smooth.construct.cr.smooth.spec()`:
! x has insufficient unique values to support 10 knots: reduce k.
```




```r
# Points colored by status, one global smoother:
ggplot(storms, aes(x = wind, y = pressure)) +
  geom_point(aes(color = status), alpha = 0.2) +
  geom_smooth()
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

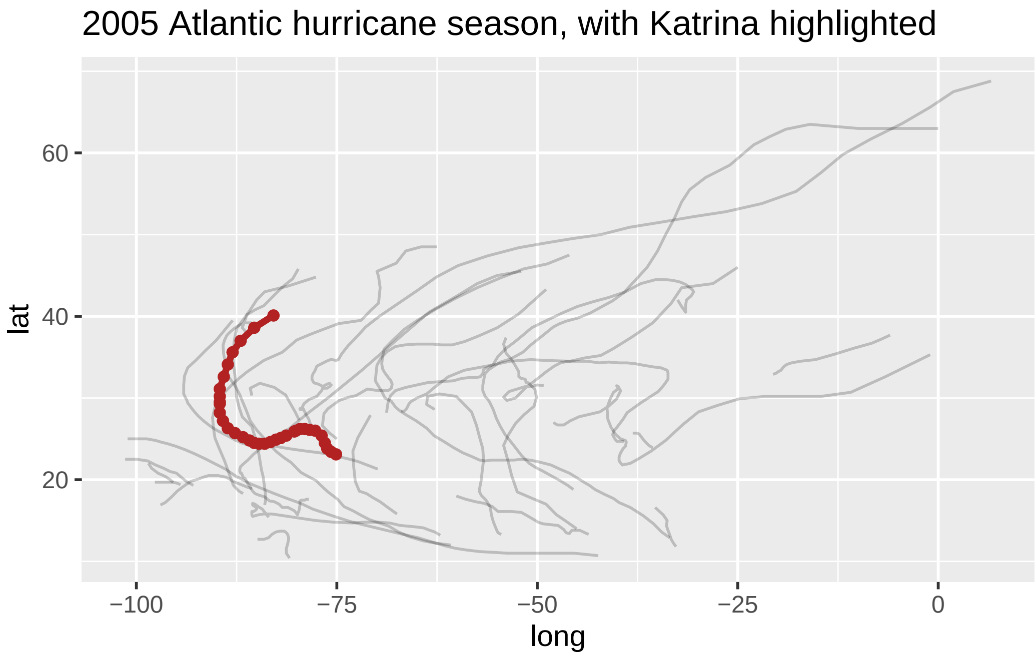


Where you put the mapping changes the plot. There's no "right" choice (just be *intentional* in your choices.)

Different data per layer

Each layer can have its **own data**. This is how you highlight a subset.

```r
katrina <- storms |> filter(name == "Katrina", year == 2005)

ggplot(storms |> filter(year == 2005),
       aes(x = long, y = lat)) +
  geom_path(aes(group = name), alpha = 0.2) +
  geom_path(data = katrina, color = "firebrick", linewidth = 1.2) +
  geom_point(data = katrina, color = "firebrick") +
  labs(title = "2005 Atlantic hurricane season, with Katrina highlighted")
```



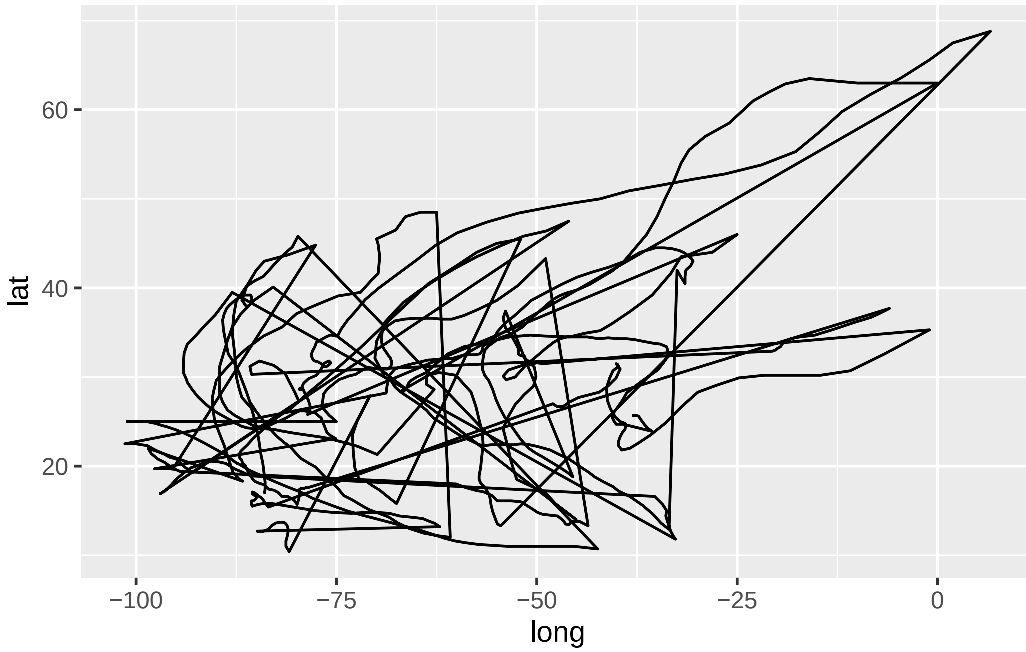
Three layers: all storms (faint), Katrina's path (red), Katrina's observations (red dots). Each calls `data =` locally to override the default.

The `group` aesthetic

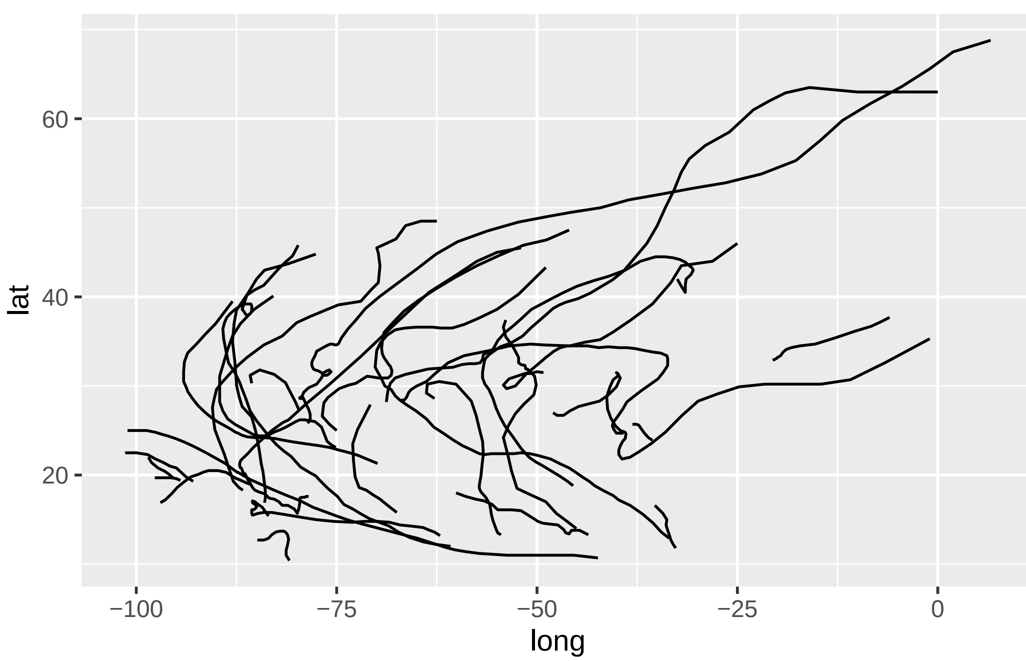
Some geoms — lines, smoothers, polygons — connect multiple rows. They need to know **which rows belong together**.

If you map a discrete variable to color or linetype, ggplot groups for you automatically. If you don't, you may need to set `group` explicitly.

```
# Bad: one giant line zigzagging between every storm
ggplot(storms |> filter(year == 2005), aes(x = long, y = lat)) +
  geom_path()
```




```r
# Good: one path per storm
ggplot(storms |> filter(year == 2005), aes(x = long, y = lat, group = name)) +
  geom_path()
```



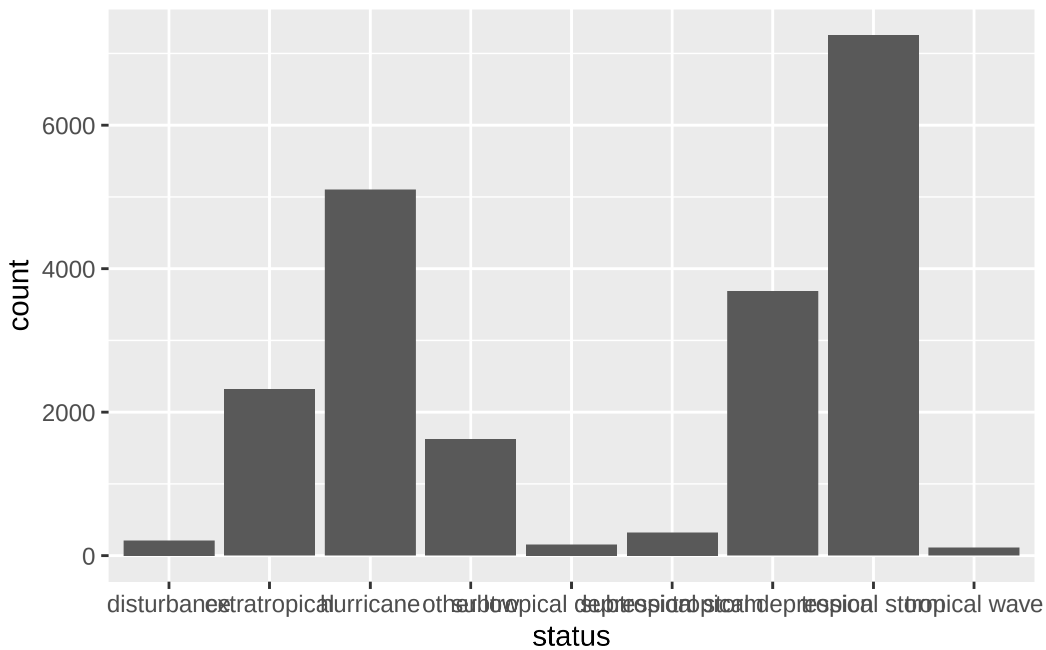
`group` doesn't add a legend or change colors, it just splits the geom.

Statistical transformations

Where do bar heights come from?

Here's a (potentially) tricky question:

```
ggplot(storms, aes(x = status)) +
  geom_bar()
```



We told ggplot about `x = status`. Where did the y-axis (count) come from?

It was computed. `geom_bar()` doesn't just plot raw data. It also *transforms* it by grouping rows by `status` and counting them, then plotting the counts.

Every geom has a **stat** (statistical transformation) it runs under the hood.

Every geom ↔ stat pair

Table 6: Geoms and their default stats.

Geom	Default stat	What it computes
`geom_bar()`	`stat_count()`	counts per x value
`geom_histogram()`	`stat_bin()`	counts per bin
`geom_boxplot()`	`stat_boxplot()`	five-number summary
`geom_smooth()`	`stat_smooth()`	fitted curve + CI
`geom_density()`	`stat_density()`	kernel density estimate
`geom_point()`	`stat_identity()`	nothing — just plots raw

You can use a geom and never think about its stat. But sometimes you will need to think about it.

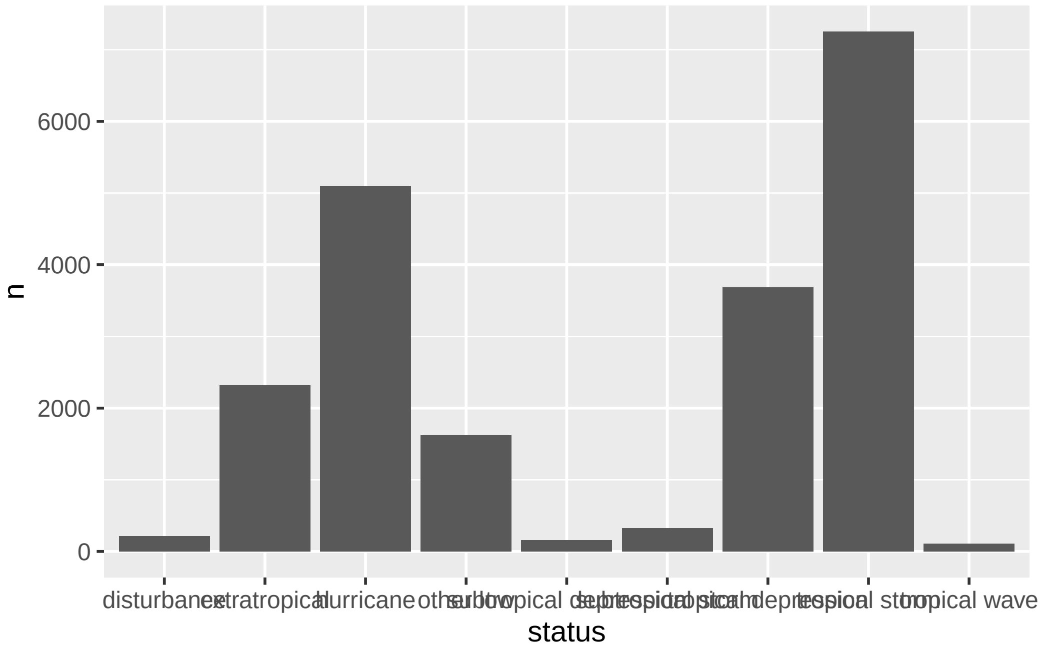
Three reasons to touch the stat

1. You already pre-computed the values. We can override the default behavior of `geom_bar` with `stat = "identity"`.

```
storm_counts <- storms |> count(status)
storm_counts
```

```
# A tibble: 9 x 2
  status                     n
  <fct>                  <int>
1 disturbance              212
2 extratropical           2318
3 hurricane               5100
4 other low               1623
5 subtropical depression   154
6 subtropical storm        323
7 tropical depression     3685
8 tropical storm          7252
9 tropical wave            111
```

```
ggplot(storm_counts, aes(x = status, y = n)) +
  geom_bar(stat = "identity")   # or just use geom_col(), it's the same thing
```



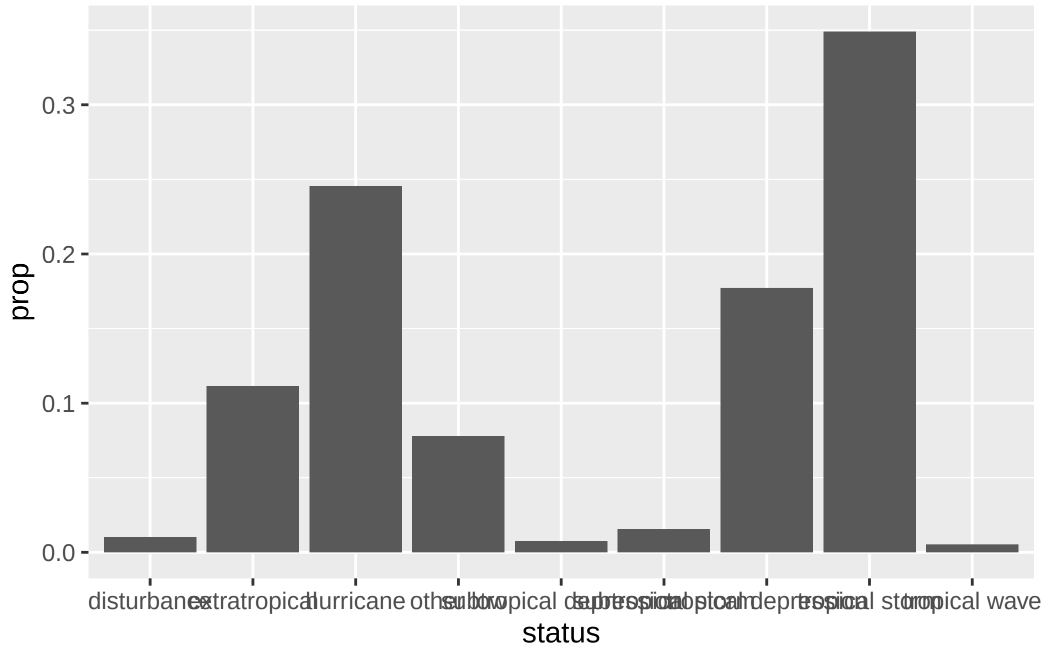
`geom_col()` is a shortcut for `geom_bar(stat = "identity")`. You can use it when you have your own y values.

Three reasons to touch the stat

2. You want a *computed* variable, not the default.

`stat_count()` actually computes two things: `count` (default) and `prop` (proportion). You can ask for the proportion using `after_stat()`:

```
ggplot(storms, aes(x = status, y = after_stat(prop), group = 1)) +
  geom_bar()
```

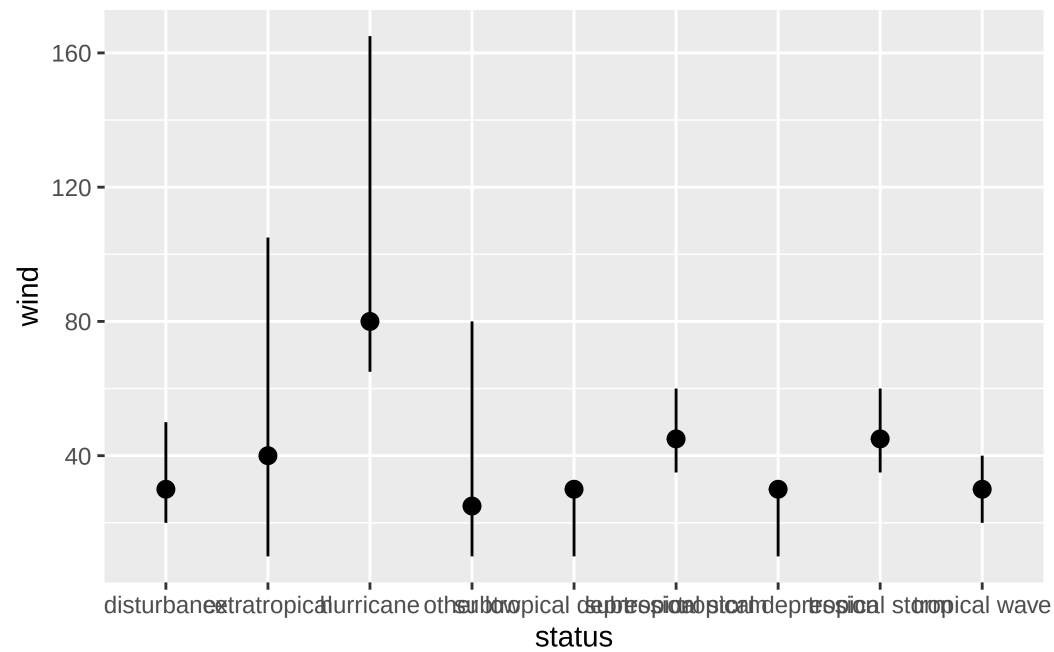


> **Why `group = 1`?**
>
> By default, `prop` is computed *within each x group*, which means every bar would be 100%. Setting `group = 1` tells ggplot "treat all rows as one group when computing proportions" so they sum to 1 across the whole plot.

Three reasons to touch the stat

3. You want to make the transformation explicit. Use a `stat_*()` function directly.

```
ggplot(storms, aes(x = status, y = wind)) +
  stat_summary(
    fun.min = min,
    fun.max = max,
    fun = median
  )
```

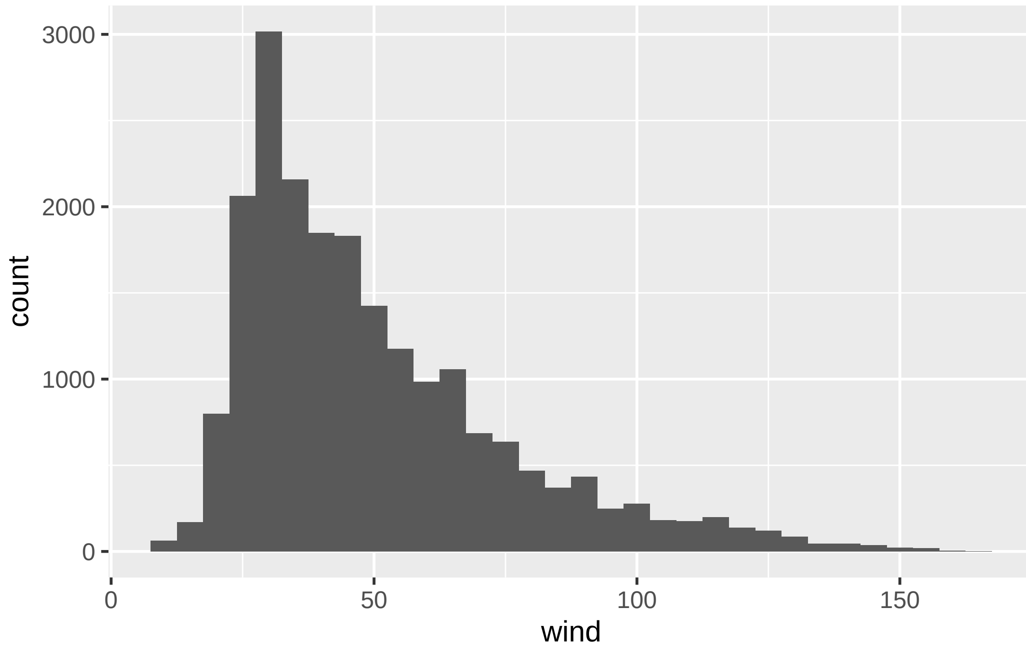


This shows the min, median, and max wind speed for each status without you having to manually write any `dplyr` code first. `stat_summary()` does the aggregation as part of the plot.

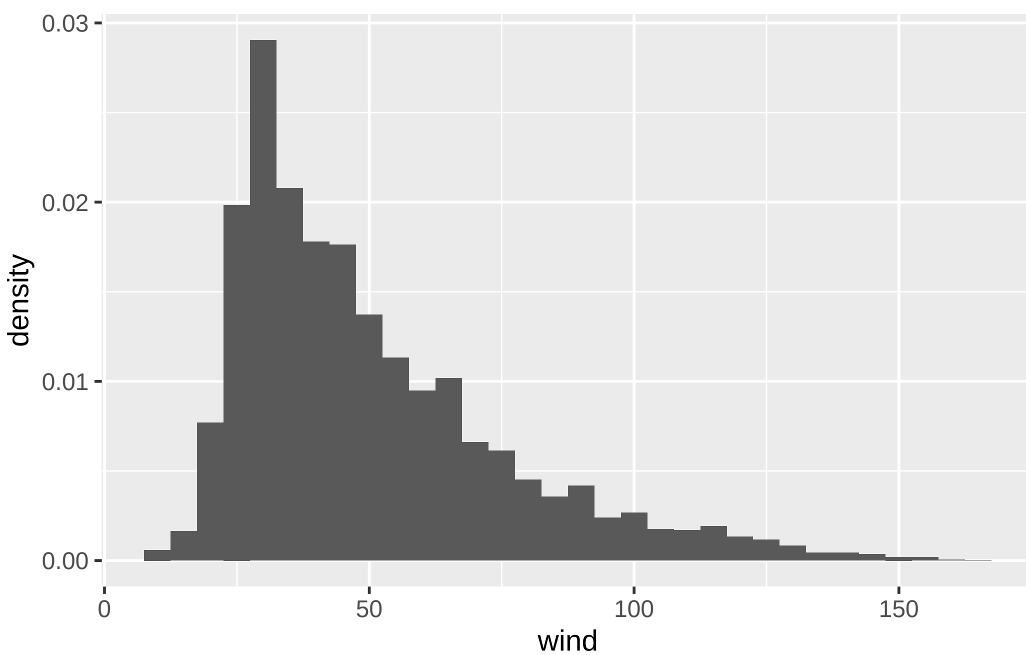
`after_stat()` quick overview

Anywhere you'd write a variable name inside `aes()`, you can wrap it in `after_stat(...)` to refer to a **computed** variable instead of one in the original data.

```
# Default: bin counts on y
ggplot(storms, aes(x = wind)) +
  geom_histogram(binwidth = 5)
```




```
# Same bins, but density on y instead
ggplot(storms, aes(x = wind, y = after_stat(density))) +
  geom_histogram(binwidth = 5)
```



To see what's available for a given geom, run `?geom_bar` and look for the **Computed variables** section.

Position adjustments

What's a position adjustment?

When multiple glyphs would land in the same spot, ggplot has to decide what to do. That decision is the **position adjustment**.

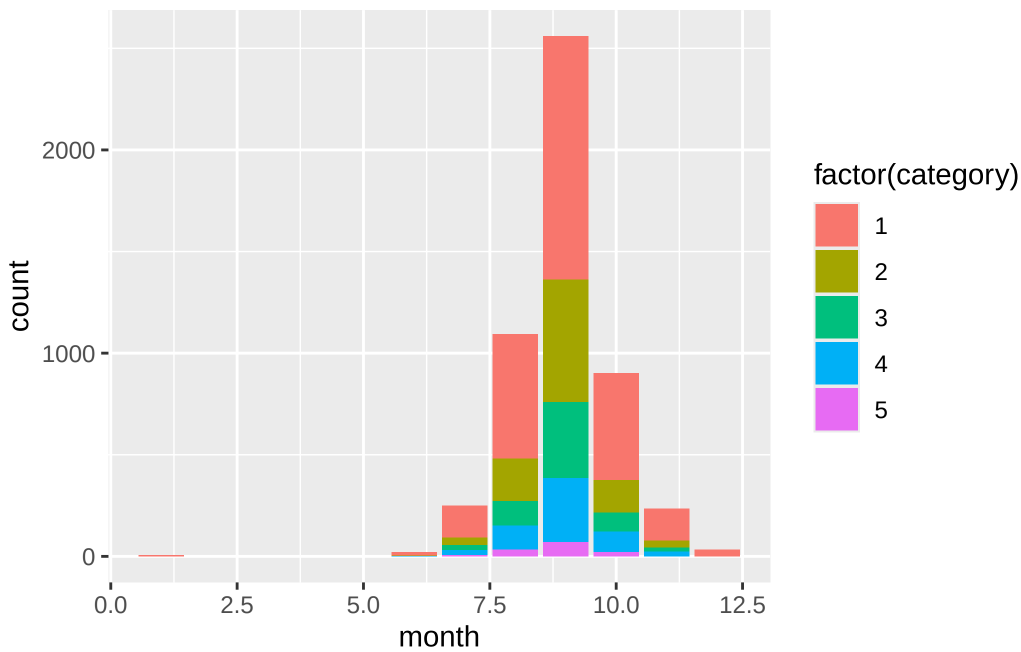
The big four:

Table 7: ggplot's main position adjustments.

Position	What it does	Default for
`"identity"`	leave it alone	points, lines
`"stack"`	stack on top of each other	bars (when filled by a 2nd var)
`"dodge"`	side-by-side	(you have to ask)
`"jitter"`	add small random noise	(you have to ask)
`"fill"`	stack and normalize to 1	(you have to ask)

Stacking (the default for filled bars)

```
hurricanes <- storms |> filter(status == "hurricane")

ggplot(hurricanes, aes(x = month, fill = factor(category))) +
  geom_bar()
```



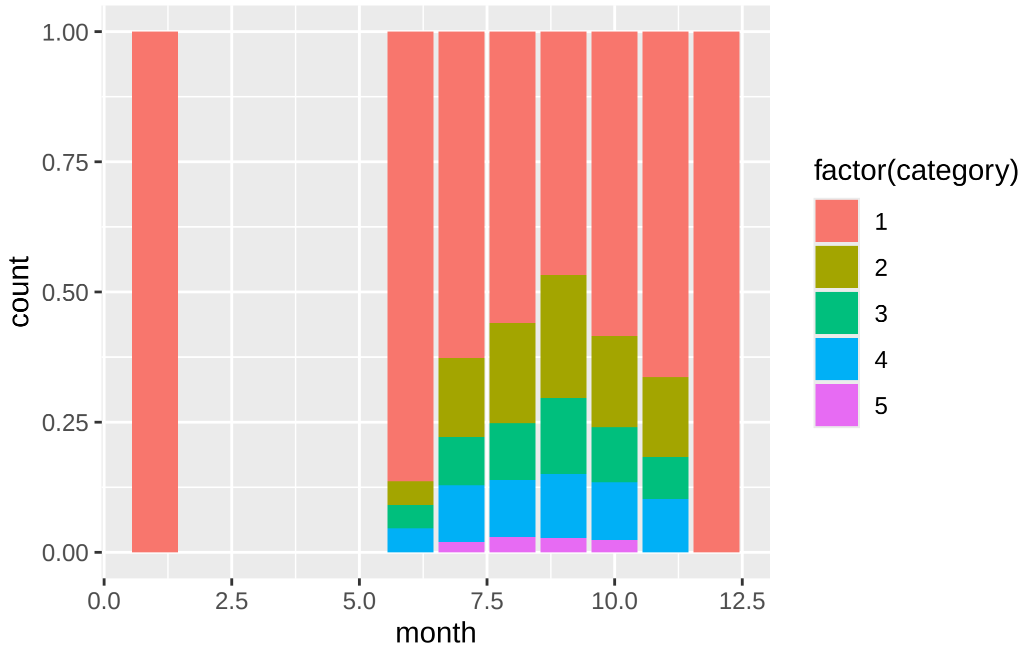
Each month gets one bar; segments stack by hurricane category.

This is useful for showing total *and* composition at once, but hard to compare middle segments across bars (their bottoms aren't aligned).

`position = "fill"`

Here we have the same code, but every bar is normalized to height 1:

```
ggplot(hurricanes, aes(x = month, fill = factor(category))) +
  geom_bar(position = "fill")
```

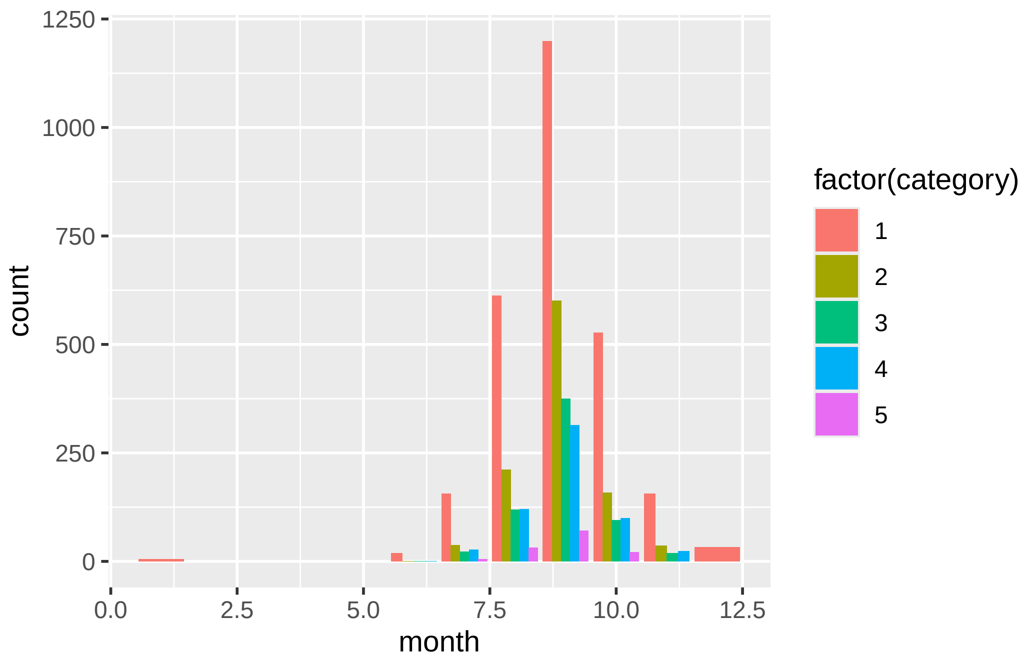


Now the y-axis is proportion, not count. Loses information about total volume, but makes the *composition* easy to compare across months.

`position = "dodge"`

Here are the bars side-by-side instead of stacked:

```
ggplot(hurricanes, aes(x = month, fill = factor(category))) +
  geom_bar(position = "dodge")
```



This is the easiest version for comparing individual counts directly. The trade-off: lots of bars, can get crowded fast.

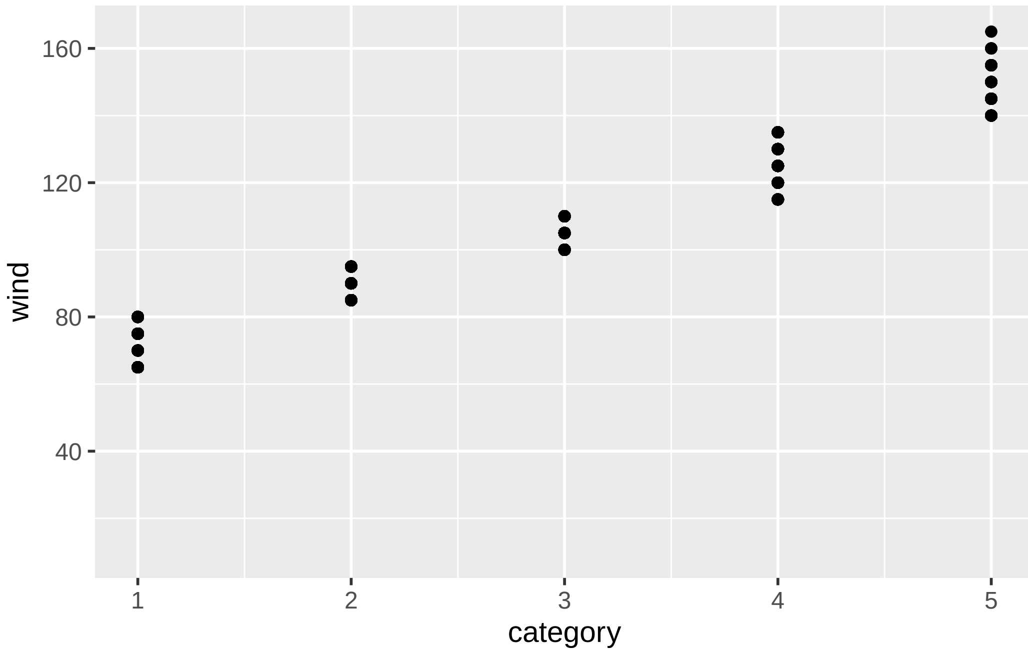
> Quick rule
>
> - **Stack:** show totals, with composition as bonus.
> - **Fill:** show composition only, ignore totals.
> - **Dodge:** compare individual values, ignore totals.

Jittering — for scatterplots, not bars

A different problem: many storm observations are recorded only every 6 hours, and pressure readings are rounded to whole millibars. Lots of points overlap.

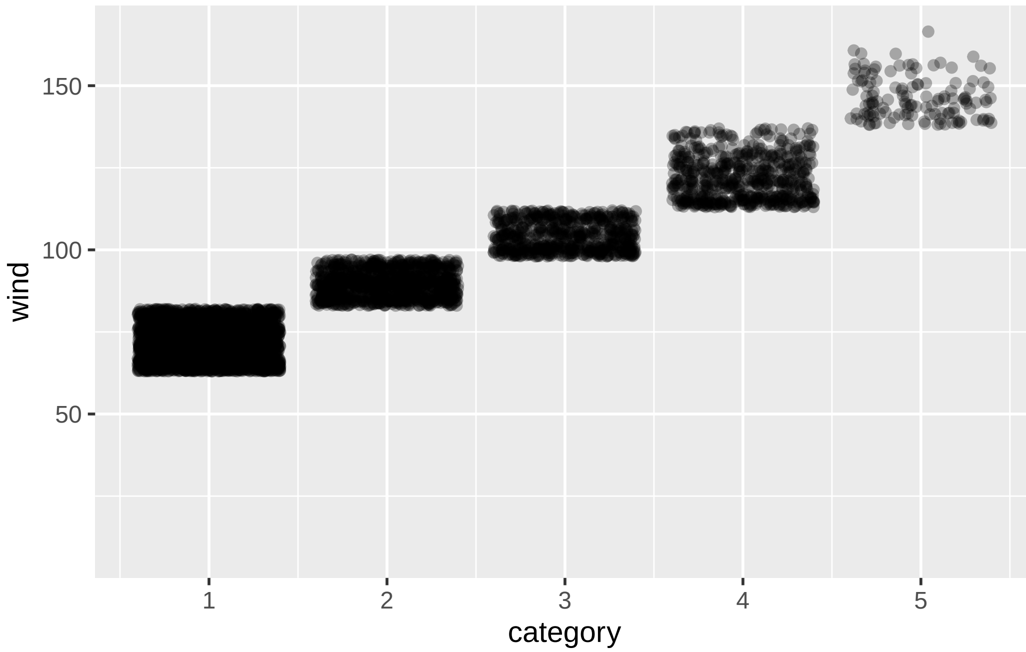
```r
# Lots of overplotting since points are stacked on top of each other
ggplot(storms, aes(x = category, y = wind)) +
  geom_point()
```

```
Warning: Removed 15678 rows containing missing values or values outside the scale range
(`geom_point()`).
```



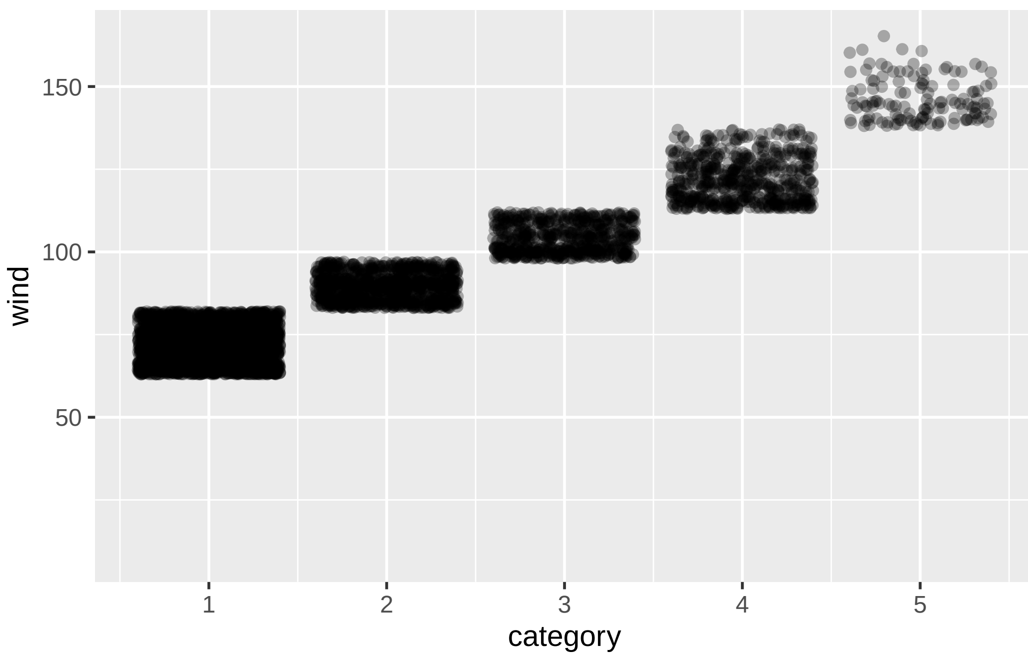

```r
# Spread them out a bit
ggplot(storms, aes(x = category, y = wind)) +
  geom_point(position = "jitter", alpha = 0.3)
```

```
Warning: Removed 15678 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
# Shorthand for the same thing
ggplot(storms, aes(x = category, y = wind)) +
  geom_jitter(alpha = 0.3)
```

```
Warning: Removed 15678 rows containing missing values or values outside the scale range
(`geom_point()`).
```



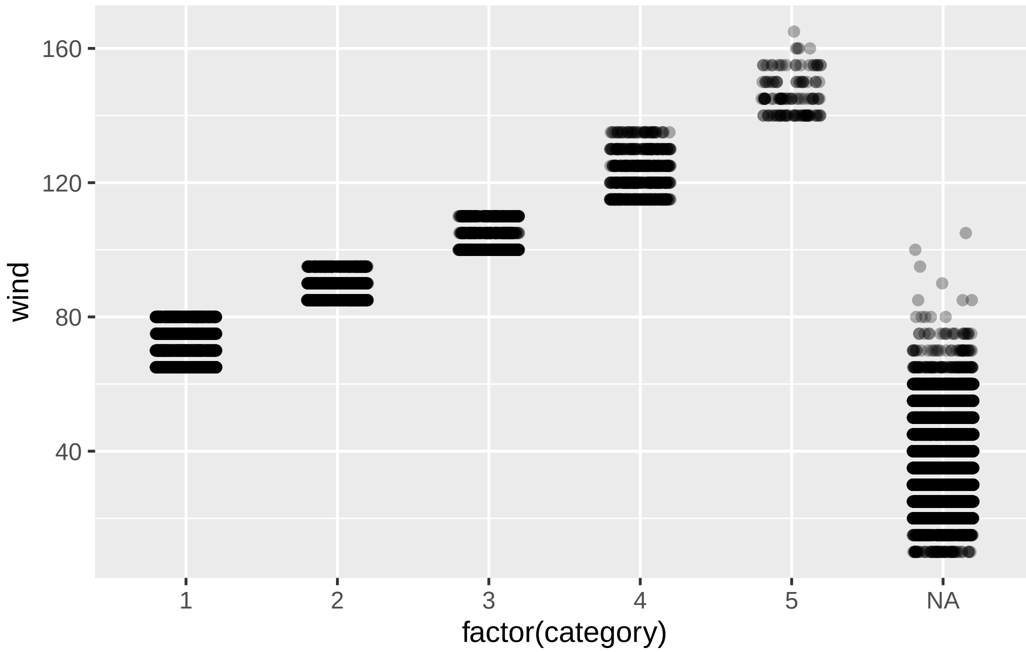
Jitter adds tiny random noise so points stop landing on top of each other. You lose precision but (often) can gain a clear picture of density.

Don't over-jitter

`geom_jitter()` takes `width` and `height` arguments. Defaults are usually fine, but:

- **Categorical x:** small `width` (~0.2), `height = 0` to keep y meaningful.
- **Integer x and y:** small jitter on both.
- **Continuous data:** you probably don't need jitter at all, just use `alpha` or `geom_hex()`/`geom_bin2d()` instead.

```
ggplot(storms, aes(x = factor(category), y = wind)) +
  geom_jitter(width = 0.2, height = 0, alpha = 0.3)
```



The big picture

The layered grammar of graphics

Putting it all together, *every* ggplot has the same skeleton:

```
ggplot(data = <DATA>) +
  <GEOM_FUNCTION>(
    mapping = aes(<MAPPINGS>),
    stat = <STAT>,
    position = <POSITION>
  ) +
  <COORDINATE_FUNCTION> +
  <FACET_FUNCTION>
```

There are seven slots here. You only want to fill the ones you need to change since ggplot picks sensible defaults for the rest.

That's why just `ggplot() + aes() + geom_*()` go a long way on their own. The other slots are there when you need them.

What we covered today

- The **vocabulary**: frame, glyph, aesthetic, scale, guide.
- **Mapping** (inside `aes()`) vs. **setting** (outside `aes()`) and the bug where you mix them up.
- Aesthetics: x, y, color, fill, size, shape, alpha, linetype, group, stroke. Discrete vs. continuous behavior.
- **Geoms**: stacking layers, global vs. local mappings, different data per layer, the `group` aesthetic.
- **Stats**: the hidden transformation in every geom; `geom_col()` vs `geom_bar()`; `after_stat()` for proportions and densities; `stat_summary()` for ad hoc aggregation.
- **Position adjustments**: identity, stack, fill, dodge, jitter — when to use which.

Activity

Activity: ggplot internals, with `storms`

Two parts.

- [Activity Canvas Link]

Part 1: Guided warmup. Five short fill-in-the-blank chunks. Practice each concept from today on a fresh question.

Part 2: Open exploration. Pick one of three guiding questions about the `storms` data and build the plot you think answers it best. We'll show a few at the end.

To turn in: Save your `.qmd` and rendered HTML.

Activity debrief

Things to watch for:

- `aes(color = "red")` instead of `color = "red"` (and getting confused when nothing turns red.)
- A `geom_path()` that zigzags wildly because `group` is missing.
- `geom_bar(stat = "identity")` when you already have counts (use `geom_col()`).
- `position = "fill"` when you actually wanted `"dodge"` (proportions vs. side-by-side counts.)
- Mapping a categorical to `size`.

Wrap-up & looking ahead

Tomorrow: facets in depth, scales, and color choices, including continuous color scales, sequential vs. diverging palettes, and accessibility.

Friday: Quiz 2 + a graphics critique activity.

Upcoming Deadlines

- **Thurs 5/28, before class**: *DC* §8.1–8.3
- **Fri 5/29, in class**: Quiz 2
- **Fri 5/29, 11:59 p.m.**: HW 2: Pipes, Plots & Reading Graphics

Reach out

Stuck? Office hours, or email me (cgc5478@psu.edu) or Xinyue (xpw5228@psu.edu).

Reference

Syllabus · AI policy · Schedule · *R4DS* Ch. 9 · *DC* Ch. 6 · ggplot2 reference